BC I/II Portfolio Specifications

The following table lists the specifications which must be met in order to earn the corresponding grade on the portfolio.

	Specifications
Excellent	❑ Portfolio is submitted as a PDF file, using the LaTeX template, and includes a table of contents. ❑ Proofs of two theorems per unit are included (12 total). ❑ Writing consistently follows most of the guidelines in the handout. ❑ Proofs are mostly complete and correct. ❑ All reflection prompts are clearly and thoughtfully answered. **Required Proofs:** ❑ A proof of the Nested Intervals Theorem and an explanation of the Bisection Method. ❑ All three "interval theorems"and the required lemmas. The Bisection Method must be used, where appropriate. ❑ The Chain Rule. ❑ The Fundamental Theorem of Calculus.
Meets Expectations	❑ Portfolio is submitted as a PDF file, using the LaTeX template, and includes a table of contents. ❑ Proofs of one theorem per unit are included (6 total). ❑ Writing makes an attempt to follow at least some of the guidelines in the handout. ❑ Proofs are mostly complete and correct. ❑ All reflection prompts are clearly and thoughtfully answered. **Required Proofs:** ❑ At least one of the three "interval theorems" and the required lemma. The Bisection Method must be used, where appropriate.

List of Theorems

Any of the following theorems may be included in the portfolio. Theorems not included on this list may be included if approved by the instructor.

The Interval Theorems

❑ Intermediate Value Theorem (Lemma: Continuous and positive at a point implies positive on an open interval)
❑ Extreme Value Theorem (Lemma: Continuous at a point implies bounded on an open interval)
❑ Integrability of a Continuous Function on a Closed Interval (Lemma: Continuous on a closed interval implies uniformly continuous)

Unit 1 - Completeness

❑ Natural numbers are not bounded above
❑ Rationals between reals
❑ Capture Theorem
❑ Flipping huge theorem
❑ Nested Intervals Theorem
❑ Bisection Method
❑ R is uncountable using the Bisection Method
❑ Rationals between reals using the Bisection Method

Unit 2 - Limits

❑ Sum, constant multiple rules for limits.
❑ Product and quotient rules for limits.
❑ Limits of powers and rational roots of x.
❑ Limits of polynomials and rational functions.
❑ Limits of trig, exponential, and logarithmic functions.

Unit 3 - Continuity

❑ Continuity of sums, constant multiples, products, quotients of continuous functions
❑ Continuity of powers and rational roots of x
❑ Continuity of composite functions
❑ Continuity of inverse functions
❑ Intermediate Value Theorem

Unit 4 - Derivatives

❑ Power rule
❑ Sums, constant multiples
❑ Product and quotient rules
❑ Chain Rule
❑ Trig functions and exponential functions
❑ Derivatives of inverse functions
❑ Inverse trig functions

Unit 5 - Applications of Derivatives

- ❑ Extreme Value Theorem
- ❑ Mean Value Theorem
- ❑ Corollaries of the Mean Value Theorem
- ❑ First Derivative Test
- ❑ Second Derivative Test
- ❑ If a continuous function has a single local extremum and no other critical points on an interval, then the local extremum is a global extremum
- ❑ L'Hopital's Rule

Unit 6 - Integration

- ❑ The Fundamental Theorem of Calculus
- ❑ Construction of the definite integral and verification of the axioms
- ❑ Linearity of the definite integral
- ❑ The Substitution Rule
- ❑ Antiderivative formulas for exponential functions and 1/x
- ❑ Antiderivative formulas for trig functions
- ❑ Integration by Parts